

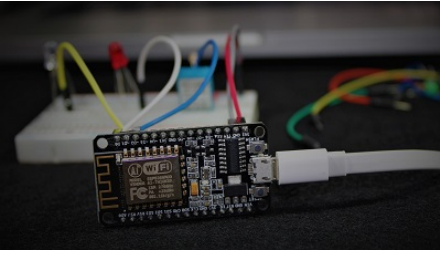
Turn Your Raspberry Pi Zero into a USB Keyboard (HID)

In this project you'll learn how to turn a Raspberry Pi Zero board into a USB keyboard or HID (Human Interface Device). After following some simple steps, you can write a Python script to make your Pi act as a USB keyboard.

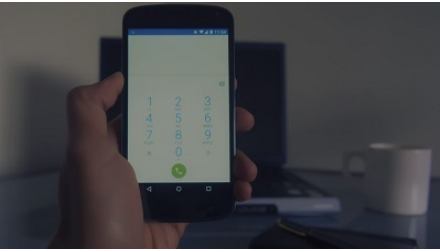
First, watch the video demonstration



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[Build Web Servers with ESP32 and ESP8266](#) » boards to control outputs and monitor sensors remotely.

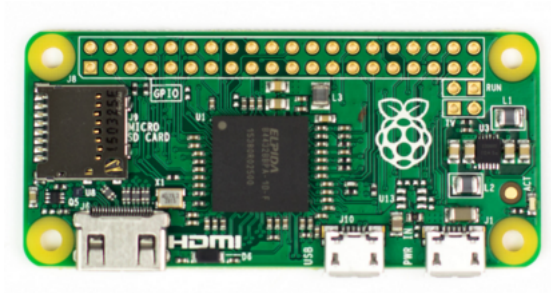
Prerequisites:

- You need a [Raspberry Pi Zero board](#)
- You should be familiar with the Raspberry Pi – [read Getting Started with Raspberry Pi](#).
- You should have the Raspbian operating system installed in your Raspberry Pi – [read Installing Raspbian Lite, Enabling and Connecting with SSH](#).

If you like home automation and you want to learn more about Node-RED, Raspberry Pi, ESP8266 and Arduino. I recommend that you download my course: [Build a Home Automation System for \\$100](#).

Parts Required

For this project you'll need a [Raspberry Pi Zero board](#). **Important:** this tutorial doesn't work with a Raspberry Pi 3 board.



1. Enabling Modules and Drivers

These next steps to prepare the Pi Zero board are based on the instructions from [iSticktoit](#). First, you need to run these three commands to enable the necessary modules and drivers:

```
pi@raspberrypi:~ $ echo "dtoverlay=dwc2" | sudo tee -a /boot/config.txt
pi@raspberrypi:~ $ echo "dwc2" | sudo tee -a /etc/modules
pi@raspberrypi:~ $ sudo echo "libcomposite" | sudo tee -a /etc/modules
```

2. Configuring the Gadget

Now, you have to define your Pi Zero (HID gadget) as a USB keyboard. The configuration is done via *ConfigFS*, a virtual file system located in */sys/*.

Creating the config script

The configuration is volatile, so it must run on each startup. Create a new file called *isticktoit_usb* in */usr/bin/* and make it executable:

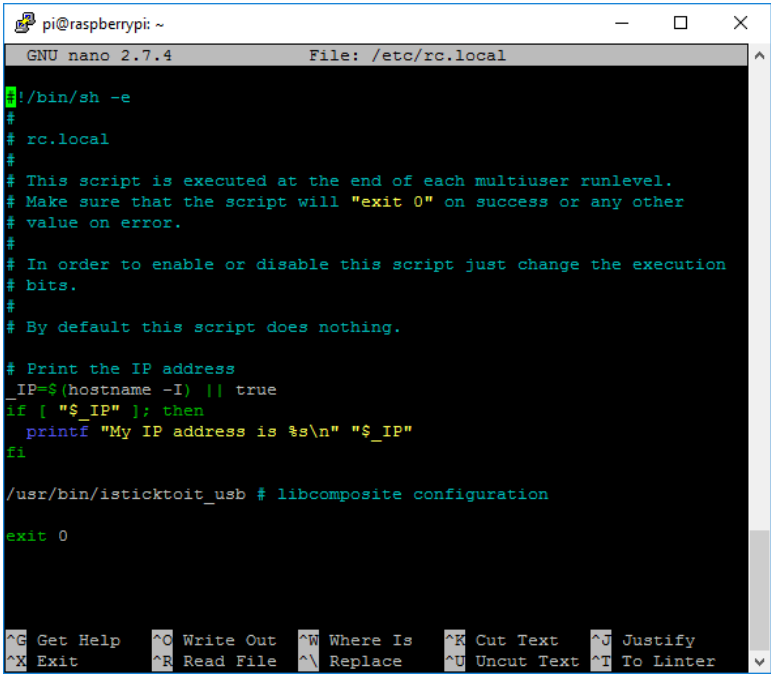
```
pi@raspberrypi:~ $ sudo touch /usr/bin/isticktoit_usb
pi@raspberrypi:~ $ sudo chmod +x /usr/bin/isticktoit_usb
```

```
pi@raspberrypi:~ $ sudo nano /etc/rc.local
```

Add the following before the line containing **exit 0**:

```
/usr/bin/isticktoit_usb # libcomposite configuration
```

Here's how your file should look like (to save the file, press Ctrl+X followed by Y and Enter):



3. Creating the gadget

For this project, we will turn the Raspberry Pi into a USB keyboard, but you could make it work as a Serial adapter, Ethernet adapter, and Mass Storage. Open the file with:

```
pi@raspberrypi:~ $ sudo nano /usr/bin/isticktoit_usb
```

Leave the default values, but you could even change the serial number, manufacturer and product name to fit your specific needs.

```
#!/bin/bash
cd /sys/kernel/config/usb_gadget/
mkdir -p isticktoit
cd isticktoit
echo 0x1d6b > idVendor # Linux Foundation
echo 0x0104 > idProduct # Multifunction Composite Gadget
echo 0x0100 > bcdDevice # v1.0.0
echo 0x0200 > bcdUSB # USB2
mkdir -p strings/0x409
echo "fedcba9876543210" > strings/0x409/serialnumber
echo "Tobias Girstmair" > strings/0x409/manufacturer
echo "iSticktoit.net USB Device" > strings/0x409/product
mkdir -p configs/c.1/strings/0x409
echo "Config 1: ECM network" > configs/c.1/strings/0x409/configuration
echo 250 > configs/c.1/MaxPower

# Add functions here
mkdir -p functions/hid.usb0
echo 1 > functions/hid.usb0/protocol
echo 1 > functions/hid.usb0/subclass
echo 8 > functions/hid.usb0/report_length
echo -ne \\x05\\x01\\x09\\x06\\xa1\\x01\\x05\\x07\\x19\\xe0\\x29\\xe7\\x15\\x06 > functions/hid.usb0/report_desc
ln -s functions/hid.usb0 configs/c.1/
# End functions

ls /sys/class/udc > UDC
```

Here's how your file should look like in the end (to save the file, press Ctrl+X followed by Y and Enter):

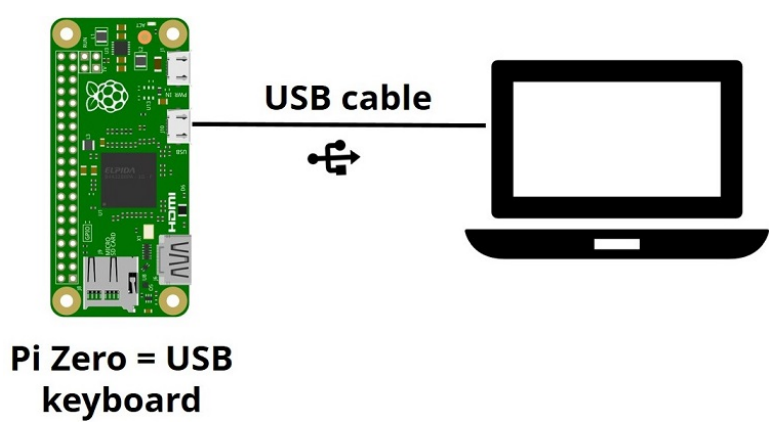
```
#!/bin/bash
cd /sys/kernel/config/usb_gadget/
mkdir -p isticktoit
cd isticktoit
echo 0x1d6b > idVendor # Linux Foundation
echo 0x0104 > idProduct # Multifunction Composite Gadget
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# Add functions here
mkdir -p functions/hid.usb0
echo 1 > functions/hid.usb0/protocol
echo 1 > functions/hid.usb0/subclass
echo 8 > functions/hid.usb0/report_length
echo -ne \\x05\\x01\\x09\\x06\\xa1\\x01\\x05\\x07\\x19\\xe0\\x29\\xe7\\x15\\x00$
ln -s functions/hid.usb0 configs/c.1/
# End functions

ls /sys/class/udc > UDC
```

4. Python Script

After preparing your Raspberry Pi Zero, connect it to a laptop or desktop computer through the micro USB port that is used for data and peripherals. That micro USB will both power the Pi Zero and act as a keyboard to the connected computer.



Establish an [SSH connection](#) with your Pi and use the next command to create a new Python script:

```
pi@raspberrypi:~ $ nano RPi_Keyboard_Example.py
```

Copy and paste the next Python script to your Raspberry Pi.

```
#!/usr/bin/env python3
NULL_CHAR = chr(0)

def write_report(report):
    with open('/dev/hidg0', 'rb+') as fd:
        fd.write(report.encode())

# Press a
write_report(NULL_CHAR*2+chr(4)+NULL_CHAR*5)
# Release keys
write_report(NULL_CHAR*8)
# Press SHIFT + a = A
write_report(chr(32)+NULL_CHAR+chr(4)+NULL_CHAR*5)

# Press b
write_report(NULL_CHAR*2+chr(5)+NULL_CHAR*5)
# Release keys
write_report(NULL_CHAR*8)
# Press SHIFT + b = B
write_report(chr(32)+NULL_CHAR+chr(5)+NULL_CHAR*5)

# Press SPACE key
write_report(NULL_CHAR*2+chr(44)+NULL_CHAR*5)

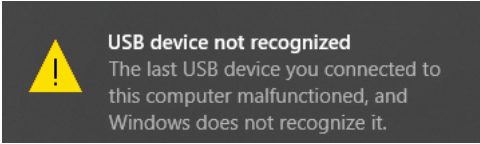
# Press c key
write_report(NULL_CHAR*2+chr(6)+NULL_CHAR*5)
# Press d key
write_report(NULL_CHAR*2+chr(7)+NULL_CHAR*5)
```

[View raw code](#)

Demonstration

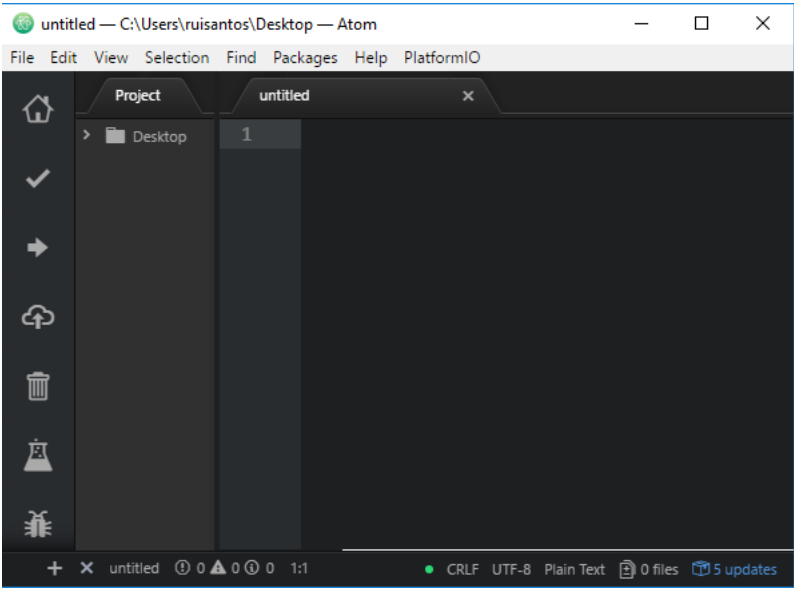
Let's test it, if you plug the Pi Zero to Computer #1, after a few seconds you'll see an alert message or sound that indicates that a keyboard was connected successfully.

without any additional configuration or drivers installation. So, you can continue and it will work just fine.



Computer #1

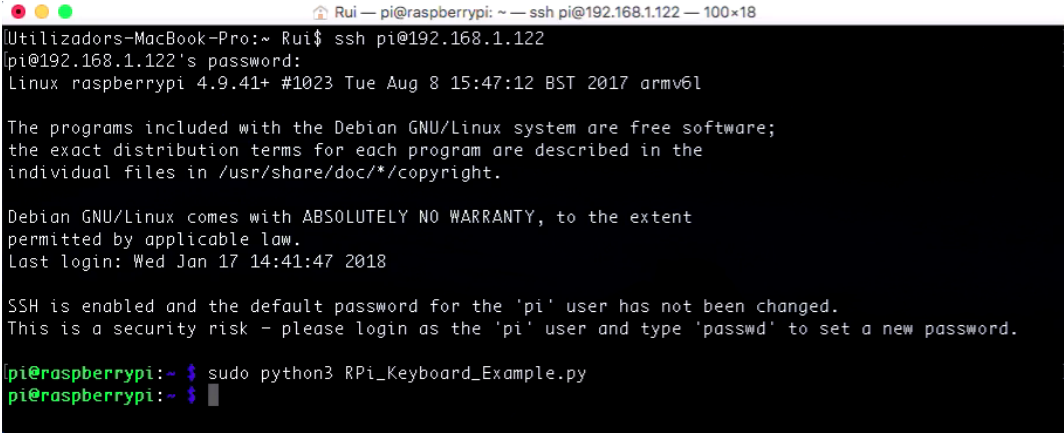
Open any text editor program and leave your cursor in the new file:



Computer #2

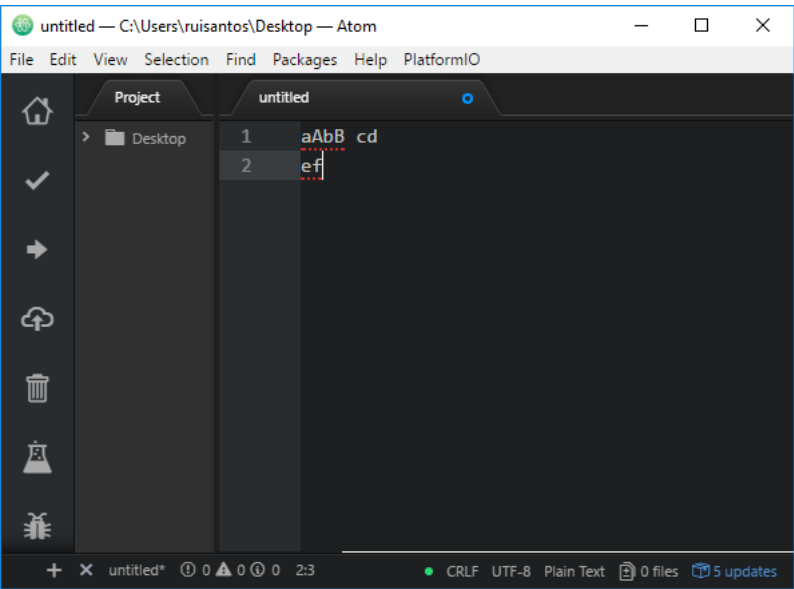
Establish an SSH connection with your Pi Zero and run the Python script created earlier:

```
pi@raspberrypi:~ $ sudo python3 RPi_Keyboard_Example.py
```



Result

The script will press these keys in that order: a – A – b – B – Space key – c – D – Enter/Return key – e – f.



You can customize the Python script to act as a keyboard and press any other character sequence.

Note: the Pi Zero also acts as a keyboard when connected to a Mac or Linux machine without any additional changes.

Taking It Further

You can use Table 12: Keyboard/Keypad Page from this [USB HID PDF](#) to find the ID of each key that you would assign in the Python script.

Table 12: Keyboard/Keypad Page

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC- Mac AT	UNI X	Boot
0	00	Reserved (no event indicated) ⁹	N/A	✓	✓	4/101/104
1	01	Keyboard ErrorRollOver ⁹	N/A	✓	✓	4/101/104
2	02	Keyboard POSTFail ⁹	N/A	✓	✓	4/101/104
3	03	Keyboard ErrorUndefined ⁹	N/A	✓	✓	4/101/104
4	04	Keyboard a and A ⁴	31	✓	✓	4/101/104
5	05	Keyboard b and B	50	✓	✓	4/101/104
6	06	Keyboard c and C ⁴	48	✓	✓	4/101/104
7	07	Keyboard d and D	33	✓	✓	4/101/104
8	08	Keyboard e and E	19	✓	✓	4/101/104
9	09	Keyboard f and F	34	✓	✓	4/101/104
10	0A	Keyboard g and G	35	✓	✓	4/101/104
11	0B	Keyboard h and H	36	✓	✓	4/101/104
12	0C	Keyboard i and I	24	✓	✓	4/101/104
13	0D	Keyboard j and J	37	✓	✓	4/101/104
14	0E	Keyboard k and K	38	✓	✓	4/101/104
15	0F	Keyboard l and L	39	✓	✓	4/101/104
16	10	Keyboard m and M ⁴	52	✓	✓	4/101/104
17	11	Keyboard n and N	51	✓	✓	4/101/104
18	12	Keyboard o and O ⁴	25	✓	✓	4/101/104
19	13	Keyboard p and P ⁴	26	✓	✓	4/101/104
20	14	Keyboard q and Q ⁴	17	✓	✓	4/101/104

For example, if you change the number highlighted in red, you can simulate a different key press:

```
write_report(NULL_CHAR*2+chr(4)+NULL_CHAR*5)
```

The number **4** correspondes to keyboard key **a**. You can find in the Usage ID (Dec) column the numbers for your desired key press sequence. If you use number **5** it corresponds to **b**, and so on...

Wrapping Up

You can use this method to make the Raspberry Pi Zero act as password filler or use it as a keystroke injection tool. That way you can easily create programs that type hundreds of keystrokes per minute.

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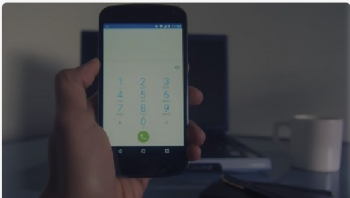




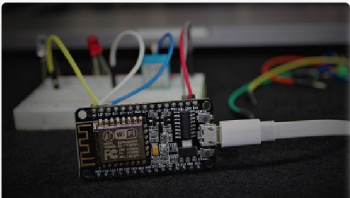
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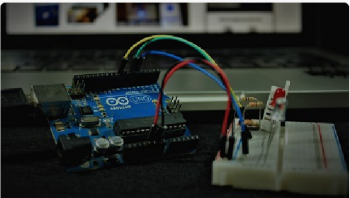
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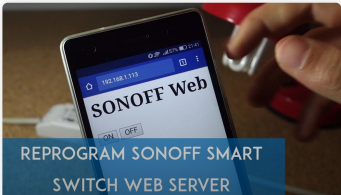


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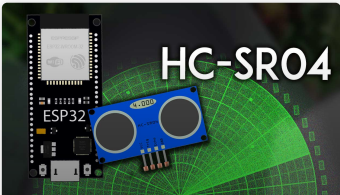


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70 thoughts on “Turn Your Raspberry Pi Zero into a USB Keyboard (HID)”

 **Duncan Amos**
January 18, 2018 at 1:49 pm

Maybe I'm missing something, but why would you ever want to do this? (serious question)

I never followed the thinking with the Arduino Leonardo either...

[Reply](#)

 **Sara Santos**
January 18, 2018 at 2:42 pm

You can use this method to make the Raspberry Pi Zero act as password filler or use it as a keystroke injection tool. That way you can easily create programs that type hundreds of keystrokes per minute.

[Reply](#)

 **Duncan Amos**
January 18, 2018 at 4:01 pm

Yes, it said that in the notes – but, to me, it seems very impractical (a button for each password you use, or remembering a number to type on another keyboard). Personally, it seems to be an example of ‘because you can, that doesn’t mean you should’, although I can see that hackers might want it.

[Reply](#)

 **Pseudonymous Bosch**
September 29, 2018 at 3:05 pm

Imagine a headless remote computer with boot password, OPAL hardware encrypted drive, (inclusive-)or LUKS software encryption on your root drive and a power outage or other unplanned reboot. This situation requires a keyboard entering passwords.

I came here looking for a possible solution to this kind of problem.

A more complicated version of this could work. Will it compromise the security more than we’re willing to risk? Depends on how we build it. I am glad that this was posted.

[Reply](#)

 **Andrew Cochrane**
April 2, 2022 at 8:23 am

arrangements. Many modern higher end keyboards don't comply with some aspect of the EDID (ASUS I'm looking at you). From what I've read usb keyboards have a limited number of "inputs" and one way of getting around it is to make the keyboard a hub with multiple keyboards attached. This doesn't work with kvms as they need to "see" a keyboard.

Reply



Robert Walk
February 20, 2023 at 1:45 pm

This is an interesting solution for a program I am going to work on. My dad has an old 400 cd player for his stereo system and you can program it with an IBM Keyboard. Using this as a base I might be able to automate most of the keystrokes needed to program the disk names into his system.

Reply



Jason Antwi-Appah
June 17, 2019 at 3:36 am

I have a work laptop on a domain, so I have to hit CTRL+ALT+DELETE, and type in my username and password. I'm using a fingerprint library so I can tell my Pi to emulate those keystrokes when the right fingerprint is used.

Reply



Van V
September 17, 2021 at 6:26 pm

Came here looking for an easy way to emulate a Mute/Unmute button for my Microsoft Teams meetings. There is an example using an Arduino CPX, but not willing to spend \$20 on that board. \$5 for a PiZero is reasonable and I happen to have a few of those laying around.

Need to wire to a momentary switch to activate the script.

Reply



cprima
September 27, 2021 at 7:34 pm

I am also here for the MS Teams shortcut!
But can you believe: The latest update *removed* the shortcut! Hopefully the next one will bring it back.
There is also a push-to-talk on the roadmap with Ctrl spacebar 😊

I am planning to write a REST API so I don't need a button and can use a StreamDeck "macrokeyboard" to trigger shortcuts (and more).

Reply



jon
January 18, 2018 at 3:01 pm

I've used this type of thing to build system test rigs – for instance to send a stream of keystrokes – with relevant delays – to automate regression testing of software.

On a more fun note I also use with my CodeClub so that they can control their scratch games with strange controllers – a bit like the MakeyMakey. <https://makeymakey.com/>

Reply

 January 18, 2018 at 4:56 pm

Hi Jon.
Thanks for sharing those interesting applications 😊

[Reply](#)

 **Duncan Amos**
January 18, 2018 at 5:07 pm

Thanks Jon

The first part makes perfect sense. The second part went completely over my head – I’m old, it happens!

[Reply](#)

 **Mike**
January 19, 2018 at 12:03 am

What Pi’s can be used if you do not have a zero? Is Pi3 too fast?

[Reply](#)

 **Rui Santos**
March 14, 2018 at 6:29 pm

At the moment, te PiZero is the only board that works with this example.

[Reply](#)

 **Ben**
February 24, 2018 at 10:23 pm

Thanks, for the amazing tutorial, helped heaps! I’m not particularly good at Python, and I was wondering if you could tell me how to get started in creating a repeater, so that when keys on the Pi’s keyboard are pressed, it sends them to the computer it is connected to?

[Reply](#)

 **Rui Santos**
March 14, 2018 at 6:16 pm

Hi Ben,
I don’t have any tutorials on that subject...
Regards,
Rui

[Reply](#)

 **David**
March 7, 2018 at 2:14 pm

Great post, but there is a small error in the instructions, in section 3, the line:
echo -ne
\\x05\\x01\\x09\\x06\\xa1\\x01\\x05\\x07\\x19\\xe0\\x29\\xe7\\x15\\x00\\x25\\x01\\x75\\x01\\x95\\x08\\x81\\\$

should read:

echo -ne
\\x05\\x01\\x09\\x06\\xa1\\x01\\x05\\x07\\x19\\xe0\\x29\\xe7\\x15\\x00\\x25\\x01\\x75\\x01\\x95\\x08\\x81\\x02\\x95\\x01\\x75\\x08\\x81\\x03\\x95\\x05\\x75\\x01\\x05\\x08\\x19\\x01\\x29\\x

(see the screenshot and the linked article from isticktoit).

Reply



Rui Santos

March 14, 2018 at 6:20 pm

You're right David.

I've fixed it! Thanks for letting me know.

Regards,
Rui

Reply



Jason Floyd

April 13, 2018 at 12:55 am

This doesn't appear fixed in your post yet. Thanks to David for the comment/fix.

Reply



Sara Santos

April 14, 2018 at 9:35 am

You're right Jason.
Thanks for letting us know.
It is fixed now! 😊

Reply



Ross

April 14, 2018 at 8:26 am

Great tutorial. Worked first time, well after I saw David's comment re the typo, which by the way is not fixed in the instructions :). But again, great stuff and thanks so much.

Reply



Sara Santos

April 14, 2018 at 9:29 am

Hi Ross.
Yes, you are right.
Everything should be fixed now.
Thanks for letting us know.
Regards,
Sara 😊

Reply



Louise

April 23, 2018 at 1:36 pm

Can't I use the PWR USB to do so?

Reply



Sara Santos

May 13, 2018 at 10:08 am

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Q

You need to use the micro USB port that is used for data and peripherals.

Reply



Joey

May 24, 2018 at 12:27 am

Do you have example for emulating mouse? Thanks

Reply



Rui Santos

June 11, 2018 at 4:53 pm

Hello Joey. Unfortunately I don't have any examples for a mouse. Thanks for the suggestion.

Regards,

Rui

Reply



steveprior

February 1, 2019 at 4:20 pm

Based on the above keyboard version, this worked for me connected as a USB mouse to Windows 7, 10 and a Raspberry Pi 3B+.

It didn't work as a mouse connected to a Pi Zero W.

In all circumstances this would only work if the device was properly recognised in Control Panel or with lsusb.

Note: I used a RPiZeroW with kernel 2017-02-16-raspbian-jessie-lite (which is version 4.4.48).

Post 4.4.50 this doesn't work.

I tested this with `sudo hid_gadget_test /dev/hidg0 mouse`

10, 10 eg moves the mouse right 10 and down 10

-b1 eg simulates a left-button click

```
#!/bin/bash
cd /sys/kernel/config/usb_gadget/
mkdir -p isticktoit
cd isticktoit
echo 0x1d6b > idVendor # Linux Foundation
echo 0x1001 > idProduct # Mouse
echo 0x0100 > bcdDevice # v1.0.0
echo 0x0200 > bcdUSB # USB2
mkdir -p strings/0x409
echo "fedcba9876543210" > strings/0x409/serialnumber
echo "Tobias Girstmair" > strings/0x409/manufacturer
echo "iSticktoit.net USB Device" > strings/0x409/product
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\x05\x01\x09\x02\xa1\x01\x09\x01\xa1\x00\x05\x09\x19\x01\x29\x03\x15\x
x00\x25\x01\x95\x03\x75\x01\x81\x02\x95\x01\x75\x05\x81\x03\x05\x01\x
09\x30\x09\x31\x15\x81\x25\x7f\x75\x08\x95\x02\x81\x06\xc0\xc0 >
functions/hid.usb0/report_desc
ln -s functions/hid.usb0 configs/c.1/
# End functions

ls /sys/class/udc > UDC
```

Reply

https://randomnerdtutorials.com/raspberry-pi-zero-usb-keyboard-hid/

10/19

July 30, 2019 at 9:29 pm

I am trying to emulate mouse movements with a raspberry pi. Could you make a video or give more instructions on the python code used to emulate mouse movements.

Reply



xmetrix
June 5, 2018 at 9:57 pm

doesn't work with raspberry pi zero w. =(

Reply



Rui Santos
June 11, 2018 at 4:23 pm

I've tried this with a RPi Zero W and it worked fine for me. Can you double-check all the steps or provide extra information that might allow me to help you?

Reply



Pseudonymous Bosch
October 1, 2018 at 11:05 am

The problem may be the version of Raspbian used. Per aidantwoods (<https://github.com/aidantwoods/RPi0w-keyboard/issues/1>) ::

“Looks like the latest Jessie now includes the broken kernel by default, which is unfortunate.

Last known one that will have a working kernel without needing to downgrade is here:
https://downloads.raspberrypi.org/raspbian_lite/images/raspbian_lite-2017-04-10/

I think you may have to avoid doing apt-get upgrade though, or the Pi will update the kernel if I recall :/”

I haven't tried with Stretch yet (wanted to make my stuff work on a known good version. Had to install python3 in addition to the using the above raspbian image. DO do an apt update but DO NOT do an apt upgrade or any variant thereof to make the python install work as of this writing.)

Reply



Sara Santos
October 3, 2018 at 9:06 am

Thank you for sharing.
We weren't aware of that issue!

Reply



Christopher D Cantrell
November 10, 2018 at 1:22 pm

Thank you for this awesome tutorial. I followed your steps on my pi zero w ... works perfectly. (I did do a “chmod +x” on the script out of habit.)

Reply



Ronnie
November 15, 2018 at 2:44 pm

I can't get it to work.

Dmsg says: (during startup)

dwc2 20980000.usb: dwc2_hstotg_enqueue_setup: failed queue (-11)

Any clue ???

Reply

 **Rui Santos**
November 15, 2018 at 5:50 pm

I'm not sure. This has been tested with the latest version of Raspbian... I'm not sure which step is failing, but if you follow these exact details it should work.

Reply

 **Ronnie**
November 16, 2018 at 3:40 pm

So the latest Kernel version: 4.14 should work???
And can Raspbian Stretch Lite be used?

<https://www.raspberrypi.org/downloads/raspbian/>

Reply

 **David Gonzalez**
December 21, 2018 at 10:36 pm

If anyone is interested I used this example and wrote my own TCP Server to send Keyboard commands over the network. Link to the code below.

<https://github.com/ddavid456/NetworkPiKeyboard>

Reply

 **Sara Santos**
December 26, 2018 at 4:42 pm

Thank you for sharing 😊

Reply

 **Cedric**
February 2, 2019 at 7:44 pm

Hi, This works fine on my Raspberry Pi 3 A+ (with a male to male usb cable) but I cannot get the g key to work..

This works: (typing f)
echo -ne "\0\0\x9\0\0\0\0" > "/dev/hidg0"
echo -ne "\0\0\0\0\0\0\0" > "/dev/hidg0"

And this works: (typing h)
echo -ne "\0\0\xa\0\0\0\0" > "/dev/hidg0"
echo -ne "\0\0\0\0\0\0\0" > "/dev/hidg0"

But this doesn't:
echo -ne "\0\0\xa\0\0\0\0" > "/dev/hidg0"
echo -ne "\0\0\0\0\0\0\0" > "/dev/hidg0"

Would be nice if someone could help.

Reply

August 26, 2020 at 5:16 am

I FOUND OUT what is the issue and HOW TO GET IT WORKING!!!!
This Works with Even the last versions of Raspberrypi Lite images.
The only issue that prevents it from working is that some of the guides out there for enabling SSH say to add “modules-load=dwc2,g_ether” to cmdline.txt.
This causes the RPI the identify to windows as an RNDIS USB Gadget which prevents it from being recognized as a USB Keyboard.
When I removed the line above from cmdline.txt and rebooted the RPI it worked!!
I think this is very important that should add it to the top of the guide, else people will spend a lot of time trying to make it work (like me).

Reply



MickT
November 1, 2020 at 2:29 am

Just thought I'd share a non-ssh work around to let you run headless and be able to revert to normal mode + easily edit and manage your python script outside the Pi.

In RPi:
– move ‘isticktoit_usb’ file to the /boot/ directory
– add your python script to the end of isticktoit_usb file “python /boot/myscript.py”
– change the ‘/etc/rc.local’ line you entered above to “/boot/isticktoit_usb # libcomposite configuration”

For raspbian the /boot/ partition is accessible to your computer when you plug the SD card into your computer. From there you can edit your python code + to boot normally just rename ‘isticktoit_usb’ to ‘anythingelseitdoesntmatter’.

Reply



Rusty
March 5, 2021 at 11:02 am

Got this working on a Pi Zero W this morning, and what I did may help others, and I'll be tesitng it on a Pi3b+ later today as well.

Output of uname -a:
pi@touchpad02:~ \$ uname -a
Linux touchpad02 5.10.17+ #1403 Mon Feb 22 11:26:13 GMT 2021 armv6l GNU/Linux

This is from a NOOB rasbain install. And it did require a couple of ‘tricks’ to get working correctly. I already had to create a .desktop file for the app I have using this setup, but even though I could run the app, it was complaining about not being authorized for /dev/hidg0. I took a look at the device via ls and found the following permissions:
pi@touchpad02:~ \$ ls -l /dev | grep hidg
crw—— 1 root root 240, 0 Mar 5 04:05 hidg0

After some trial and testing, found that changing the ownership from root:root to root:dialout didn't help, even though it did change the permissions to:
pi@touchpad02:~ \$ ls -l /dev | grep hidg
crw-rw— 1 root dialout 240, 0 Mar 5 04:05 hidg0

What ultimately worked was changing the permissions to
pi@touchpad02:~ \$ ls -l /dev | grep hidg
crw-rw-rw- 1 root root 240, 0 Mar 5 04:05 hidg0

However this isn't ‘persistent’ as the device is created at each boot. Going through the options, I tried setting the permissions in an @reboot entry in the sudo crontab file, that didn't work. What I ended up with is a ‘cleanup.desktop’ entry in ~/.config/autostart/ that looks like this:
pi@touchpad02:~ \$ cat .config/autostart/cleanup.desktop
[Desktop Entry]
Type=Application
Name=Cleanup
Exec=lxterminal -e sudo chmod 666 /dev/hidg0
pi@touchpad02:~ \$

Now the app, basically a poor-man's elgato macro screen, is able to send keyboard combinations that OBS Studio or Streamlabs OBS can recognize as hotkeys to do things like start and stop streaming, recording, toggling mute state for microphone or audio inputs, w/o affecting the app that I'm streaming.

system up with an fsoverlay to make the file system effectively read-only. (It's not like I haven't run into other pi's that have ended up with corrupt file systems.) This will protect the system even if I don't hit the shutdown button on the device before shutting down the system it's acting as a macro board for.

Reply

 **Alex**
March 17, 2021 at 10:59 am

I'm having trouble setting this up on my Raspberry Pi Zero W.

The error I'm getting is "BrokenPipeError: [Errno 108] Cannot send after transport endpoint shutdown". Is this the problem that you fixed above?

Reply

 **philipp shilov**
March 24, 2021 at 5:32 pm

Plug it with other micros port of your RPI0W.

Reply

 **luk**
July 26, 2021 at 6:22 am

You connected your pi to pwr (usb port) and you should connect to usb port (this second one)

Reply

 **Manzazuu**
August 23, 2021 at 4:02 pm

Same problem here, despite of being connected to the data usb port.
The problem was caused by the cable : I replaced the power bank usb cable I was user by a micro usb that came with the Pi and it worked perfectly.
I guess all usb cables are not equals.

Reply

 **Alexander Ehre**
December 13, 2021 at 9:28 am

Hello Rusty,

I think that I have the same problem. Could you please explain a bit more in detail what you did? It looks for me like: -made this false, -made this false... – made something could workt.. etc. So your work is for me unfortunately not clear.

Thank you!

Reply

 **Alex Eh**
December 13, 2021 at 10:30 am

Hello,

could you please explain a bit more in detail what you did?

Thanks!

 **Christian Duran**
August 2, 2021 at 4:52 pm

Could this same principle be used to make a macropad? Have a python program that says if I press this key run this program?

[Reply](#)

 **Joel Christensen**
September 30, 2021 at 9:10 pm

I plan to use it for this exact purpose

[Reply](#)

 **Tehseen**
October 5, 2021 at 12:47 am

Hi, This is a great project. I tried on My RPi Zero W and it worked perfectly.
But I want to use the KEyboard functionality over the GPIO pins of Raspberry Pi because a USB device is already connected to my Micro USB port of the Raspberry Pi Zero, so I want to emulate the Keyboard from the GPIOs and connect the GPIO to another Computer to enter the data in the Active Field at the cursor. How Can I implement this is this method?

[Reply](#)

 **Alexander Ehre**
December 13, 2021 at 9:25 am

Hello,

I've got the problem that I get these message if I start the program:

“File “/home/pi/RPi_Keyboard_Example.py”, line 14, in
write_report(NULL_CHAR2+chr(4)+NULL_CHAR5)
File “/home/pi/RPi_Keyboard_Example.py”, line 7, in write_report
with open('/dev/hidg0', 'rb+') as fd:
FileNotFoundError: [Errno 2] No such file or directory: '/dev/hidg0’”

[Reply](#)

 **Tim**
December 31, 2021 at 12:37 am

Hi Alexander. I'm running this in December 2021 as well and I'm getting the exact same error. Were you able to resolve this and get it working?

FileNotFoundError: [Errno 2] No such file or directory: '/dev/hidg0'

[Reply](#)

 **Alexander**
February 7, 2022 at 1:50 pm

Hi Tim,

Now that I have finished all the other construction sites in my project, my head is a bit freer for the HID problem and I was able to find a solution.

Rusty has already described a solution, which unfortunately was not understandable for me 2 months ago.

For this reason I would like to tell you what you have to do:

paste. This will allow the Pi to be recognised as a keyboard. By the way, it doesn't matter which Pi. I currently have it running with a Pi Model 4 B.

After that, you should use Rusty's solution. Since it is unfortunately very incomprehensible, here is a short summary without that didn't work, that did, etc.

```
$cd .config/
$mkdir autostart
$cd autostart/
$nano cleanup.desktop
[Desktop Entry]
Type=Application
Name=Cleanup
Exec=lxterminal -e sudo chmod 666 /dev/hidg0
```

Afterwards it should work. Feel free to contact me if it works or not!

Translated with <http://www.DeepL.com/Translator> (free version)

Reply



Tim Fenney
July 24, 2023 at 3:49 am

It matters which Pi.

The only ones that have the controller properly wired up, to my knowledge, are:

- zero 1/2 [w]
- 3A
- 4B (any memory)

Reply



Corpo Rate
December 15, 2021 at 8:17 pm

Just wanted to add that this works well also on the OrangePi Zero.

Thanks for posting this.

Reply



Gerald
November 26, 2022 at 3:44 am

Great tutorial! One thing I don't understand...in the python script you only send the "release keys" command after certain letters. Why doesn't it have to be called after every key press? For example why doesn't release keys have to be called right before sending "b"? Wouldn't that mean the Shift + A is still being held down?

Reply



Spinach Rick
December 31, 2022 at 1:51 pm

so how would one send two or even three key presses at the same time?

Reply



Chris
December 31, 2022 at 6:16 pm

- Send Key 1 (eg A)
- Send Key 2 (eg B)

Its explained above – look at the Python code for exmaples.

Reply



Spinach Rick
December 31, 2022 at 11:48 pm

so If i was using a joystick to emulate a keyboard, eg the joystick’s potentiometer would emulate the left/right on the keyboard. A tactile button will emulate “up” on a keyboard. Would the main look something like this?

note: x represents the x-axis of the joystick.

```
def main():  
    if x150:  
        write_report(chr(32)+NULL_CHAR+chr(79)+NULL_CHAR5)  
    if nc.buttons.Z:  
        write_report(chr(32)+NULL_CHAR+chr(79)+NULL_CHAR5)  
    else:  
        write_report(NULL_CHAR*8)
```

Reply



hasrul
January 12, 2023 at 3:41 pm

It works with me.
But how to disable this USB keyboard thing and make the USB port in raspi back to normal again?

Reply



jsmith
February 4, 2023 at 8:25 pm

This works great only problem is using this example I cannot see any way to properly send certain keys like left control key or right control key.

Left control is supposedly 224 dec and right control is 228

These keys are not sent when using 224 or 228 like in the example python code when doing a, or b, etc..

Anyone have any idea how to get left control key (224) sent from the PI?

Thanks

Reply



Laurence
February 16, 2023 at 2:45 pm

So if I wanted to use this as a mouse button remapper so my mouse can copy and paste using its keys, as there is no onboard memory, is it possible to that with this?

Reply



Jdramsey
February 28, 2023 at 7:03 pm

Have a retro keyboard (i.e Tandy 102 computer) and would like to use it on a modern PC. I bet there’s enough GPIO pins to make a keyboard encoder. Need to come up with a wiring plan and software to encode key presses. Goal is to utilize keyboard with a new display hooked to a Raspberry Pi 3 or 4 and use as a serial terminal for retro computers.

Reply

 **Sebastian Schwarzbrunn**

May 4, 2023 at 11:44 am

Replacing the USB-cable solved it for me too.

[Reply](#)

 **Jack**

July 16, 2023 at 4:33 am

If you're getting the "FileNotFoundError: [Errno 2] No such file or directory: '/dev/hidg0'" error, on a zero w, here's what worked for me:

ssh to your pi
sudo nano /usr/bin/isticktoit_usb

Once it opens, add the following below "ls /sys/class/udc > UDC"
chmod 666 /dev/hidg0

[Reply](#)

 **Stewart Grainger**

August 3, 2023 at 12:11 am

My Raspberry Pi Zero work on a Windows and Mac machine, however when I plug into my Mac Mini that is waiting for me to enter a FIRMWARE password, it will not recognize it (the keyboard) not the firmware password.

When I try to run my script :

sudo python3 RPi_Keyboard_Example2.py

It does nothing. Tried changing cables, tried rebooting.

Anyone else had that issue?

[Reply](#)

 **Kromagnon**

September 15, 2023 at 9:08 pm

Hey, I'm making a pixel bot for WoW, are sent keystrokes detected as coming from a real keyboard or as an injection? (LLMHF_INJECTED)

[Reply](#)

 **omar**

October 2, 2023 at 10:40 pm

Hello, when i try to make the rpi click the control key by doing '0xE0', it clicks the shift key, im sure this isnt just the problem for this key, any solutions?

[Reply](#)

 **Yves**

October 27, 2023 at 4:05 pm

Using this approach, how can I support the ALT keyboard codes? For example: "é" correspond to the keyboard keys "ALT + keypad_0 + keypad_2 + keypad_3 + keypad_3". More specifically, that would be "press/hold ALT while pressing/releasing keypad_0, then pressing/releasing keypad_2, then pressing/releasing keypad_3, then pressing/releasing keypad_3 and finally releasing ALT.

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